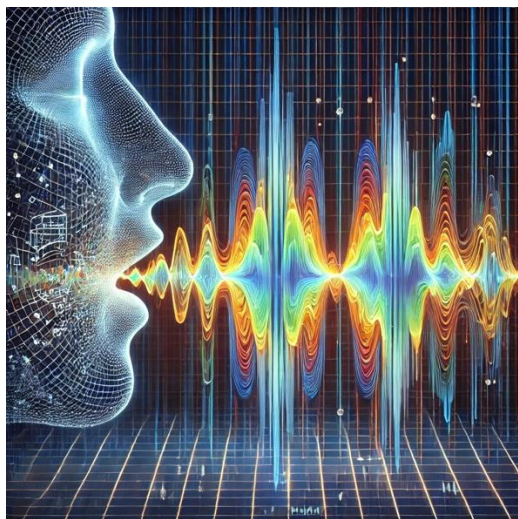




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UNIVERSITÀ DI BOLOGNA



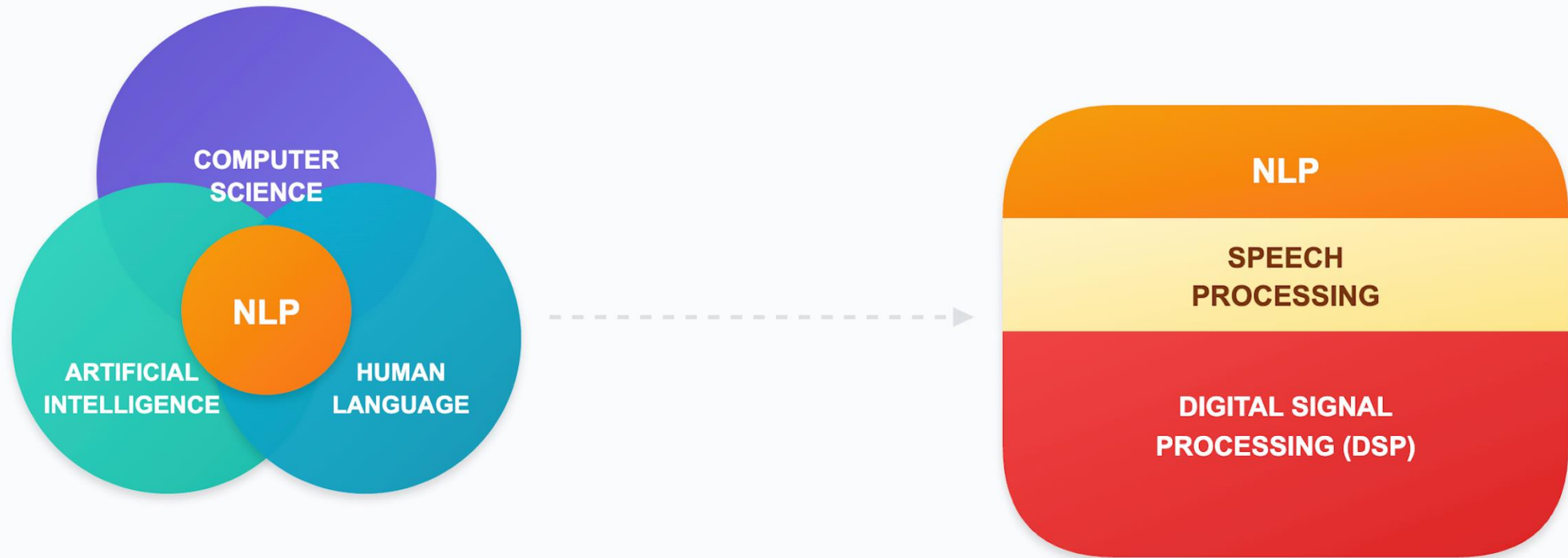
The Language of Sound: A Journey Through Speech Processing and Multimodality

Dr. Eleonora Mancini

Ph.D. Candidate at Language Technologies Lab
DISI, University of Bologna

What's Speech Processing?

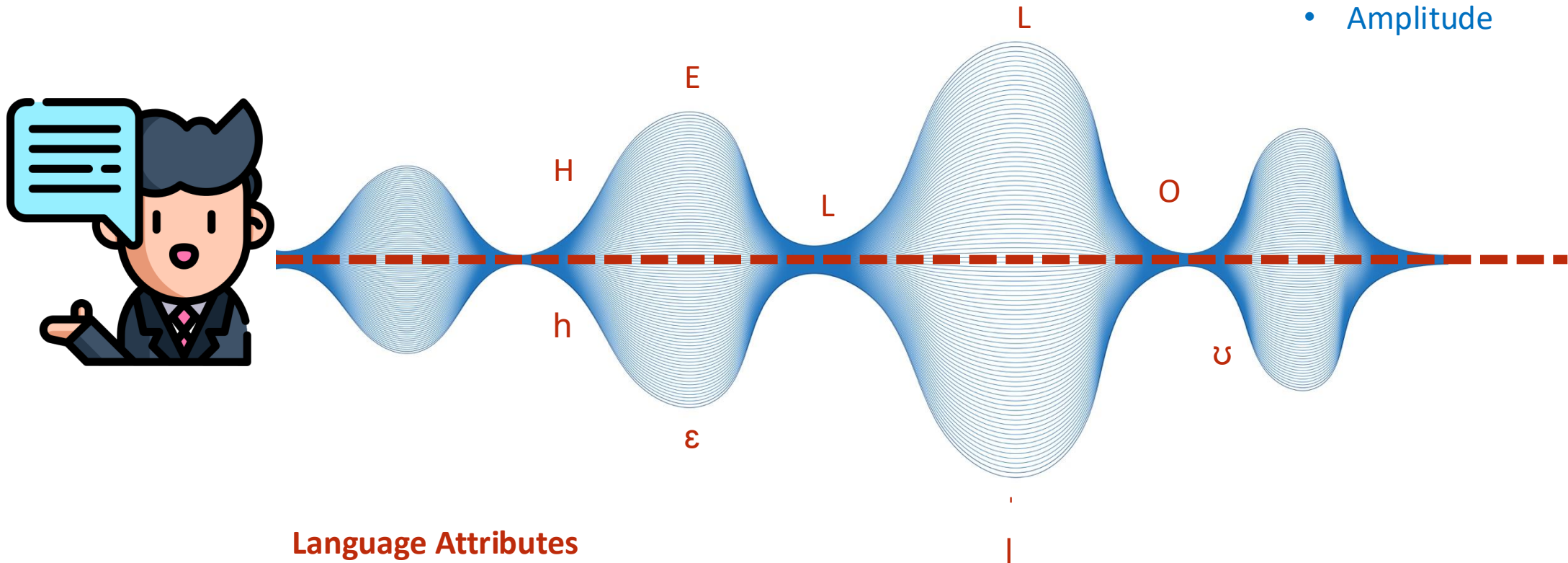
- Transforms raw sound waves into linguistic or acoustic information
- Central to speech analysis, synthesis, and recognition
- Connects physical audio signals with meaning



Speech: A Bridge Between Signal and Language

Signal Attributes

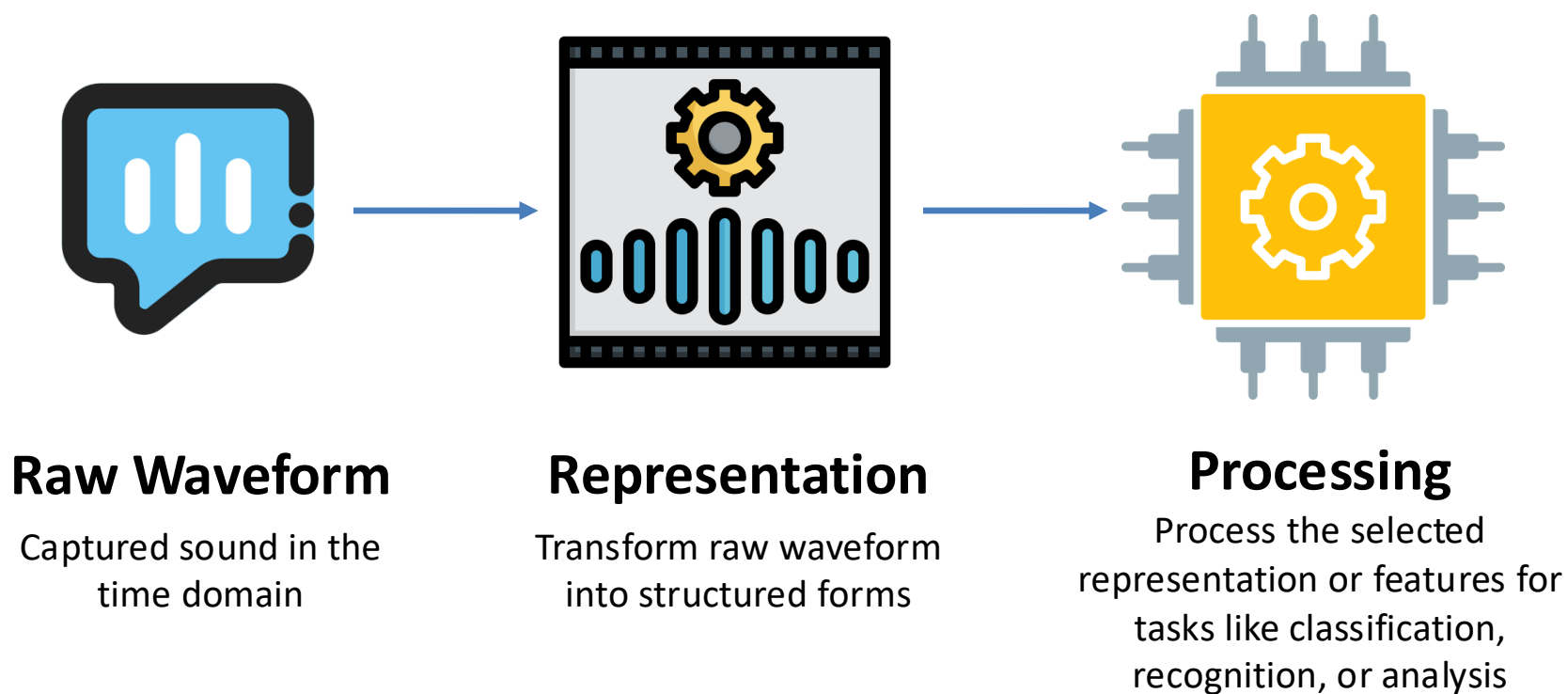
- Sound Waves
 - Frequency
 - Amplitude



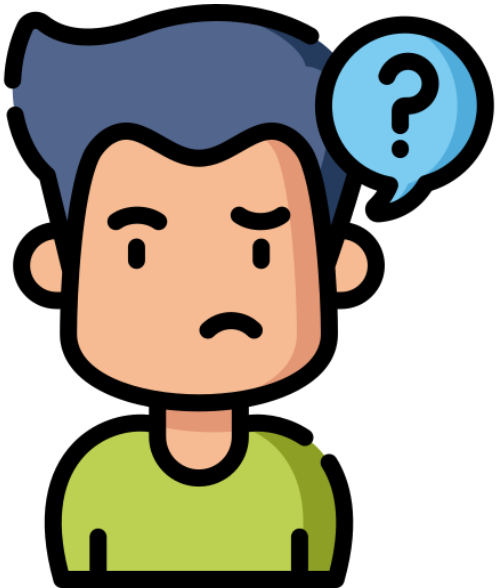
Language Attributes

- Semantic Meaning
- Phonemes
- Grammatical Structure

The Speech Processing Pipeline



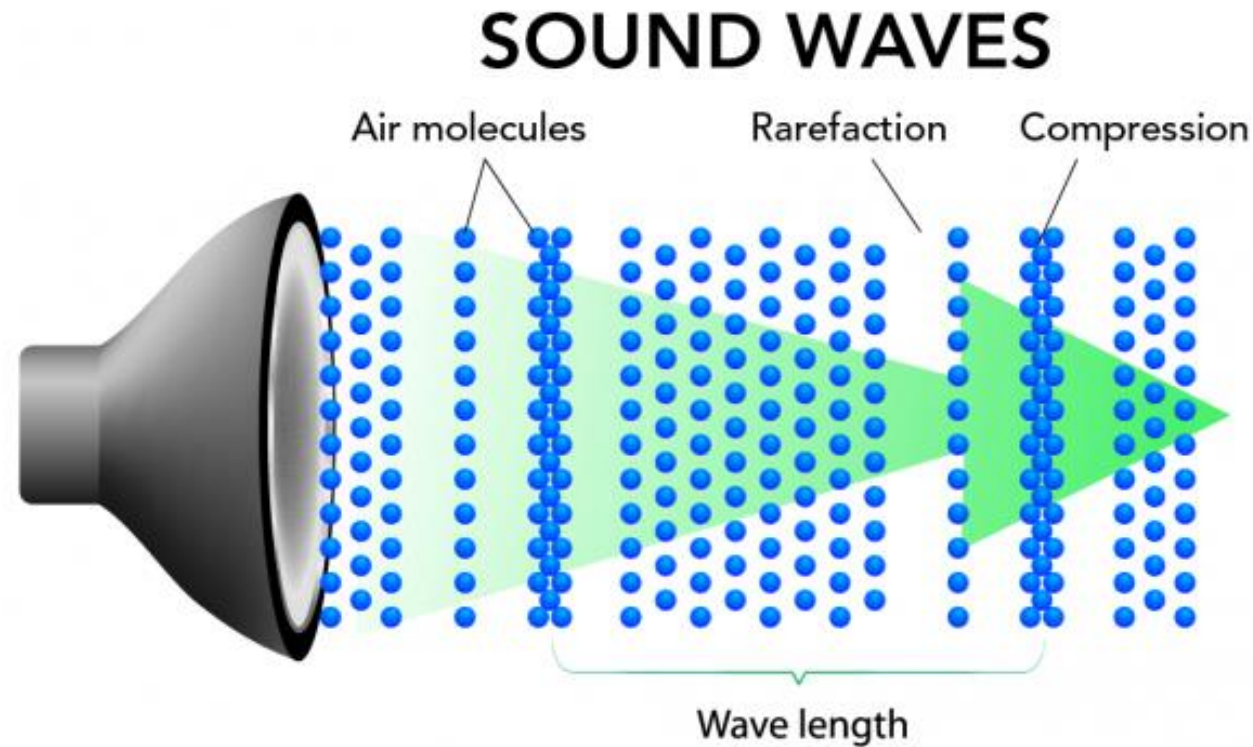
But... How do we represent the sound?



01 SPEECH (AUDIO) AS A PHYSICAL SIGNAL

Understanding Sound, Waveforms, and Signals

Sound

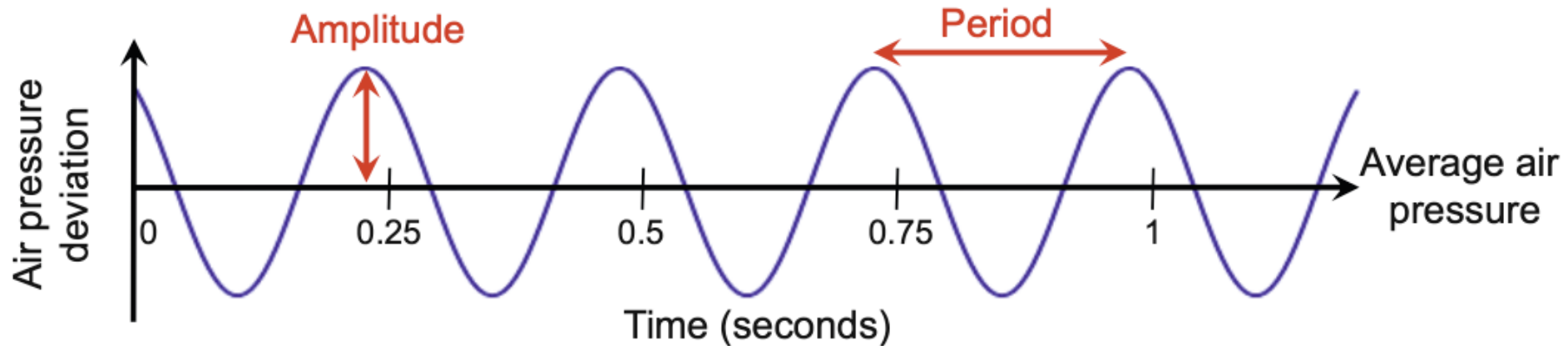


- Produced by the vibration of an object
- Vibrations determine oscillation of air molecules
- Alternation of air pressure causes a wave
- **Pitch**: frequency of the vibrations
- **Loudness**: amplitude of the vibrations

Understanding Sound, Waveforms, and Signals

Waveform: Representation of a Sound

$$y(t) = A \sin(2\pi ft + \varphi)$$



FREQUENCY

- Represents the **pitch** of the sound
(how fast the air pressure oscillates)

$$f$$

PERIOD

- Distance between two peaks
- Inversely correlated to frequency

$$f = \frac{1}{T}$$

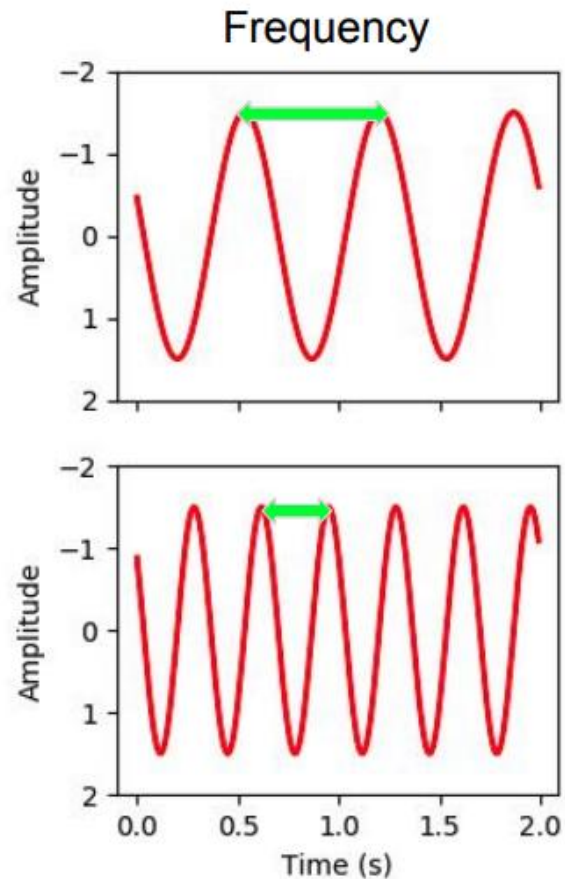
AMPLITUDE

- Measure of air pressure
- Measured from a point on the waveform to the 0 amplitude
- Represents the **loudness** of the sound.

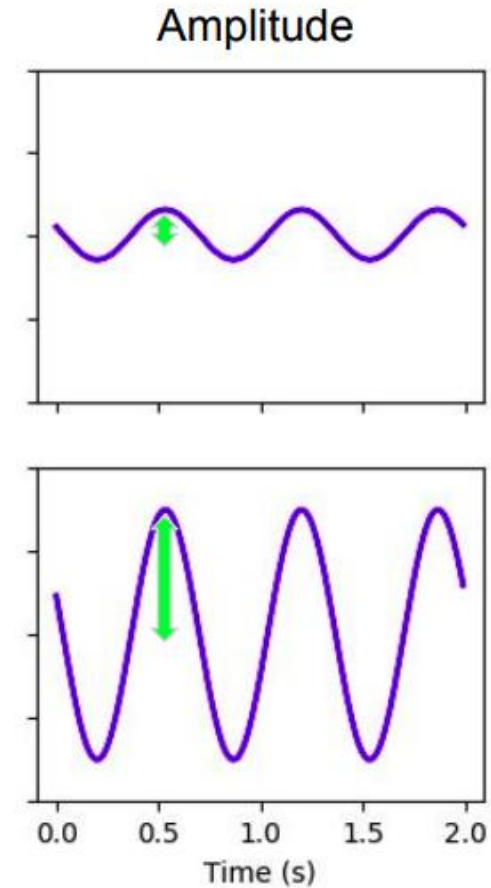


Understanding Sound, Waveforms, and Signals

Waveform: Representation of a Sound



Higher frequency -> Higher pitch



Larger amplitude -> Louder

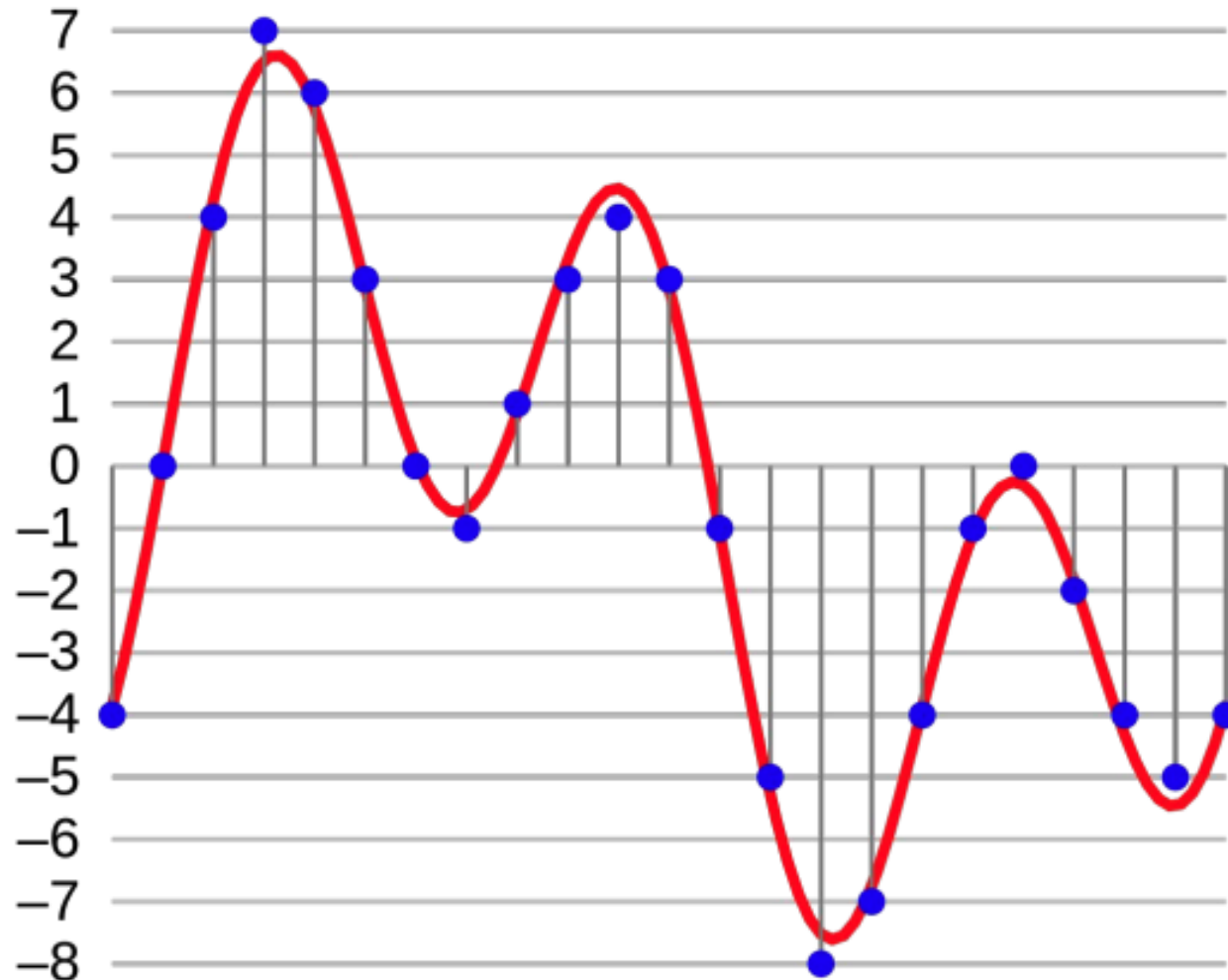
Understanding Sound, Waveforms, and Signals

From Waveform to Signal

- A **signal** is a representation of information.
- In the case of sound, the **signal** is the **waveform** that carries the audio information and changes over time (e.g. person speaking or instrument playing) which can be either continuous (**analog**) or discrete (**digital**).
 - **Analog Signal:** The continuous waveform (analog sound) as it exists in the real world
 - **Digital Signal:** The analog waveform is **sampled** (measured at specific time intervals) and **quantized** (converted to a discrete set of values), which creates a digital signal. This process allows computers and digital devices to store and manipulate sound.



Analog digital conversion (ADC)



Two main steps:

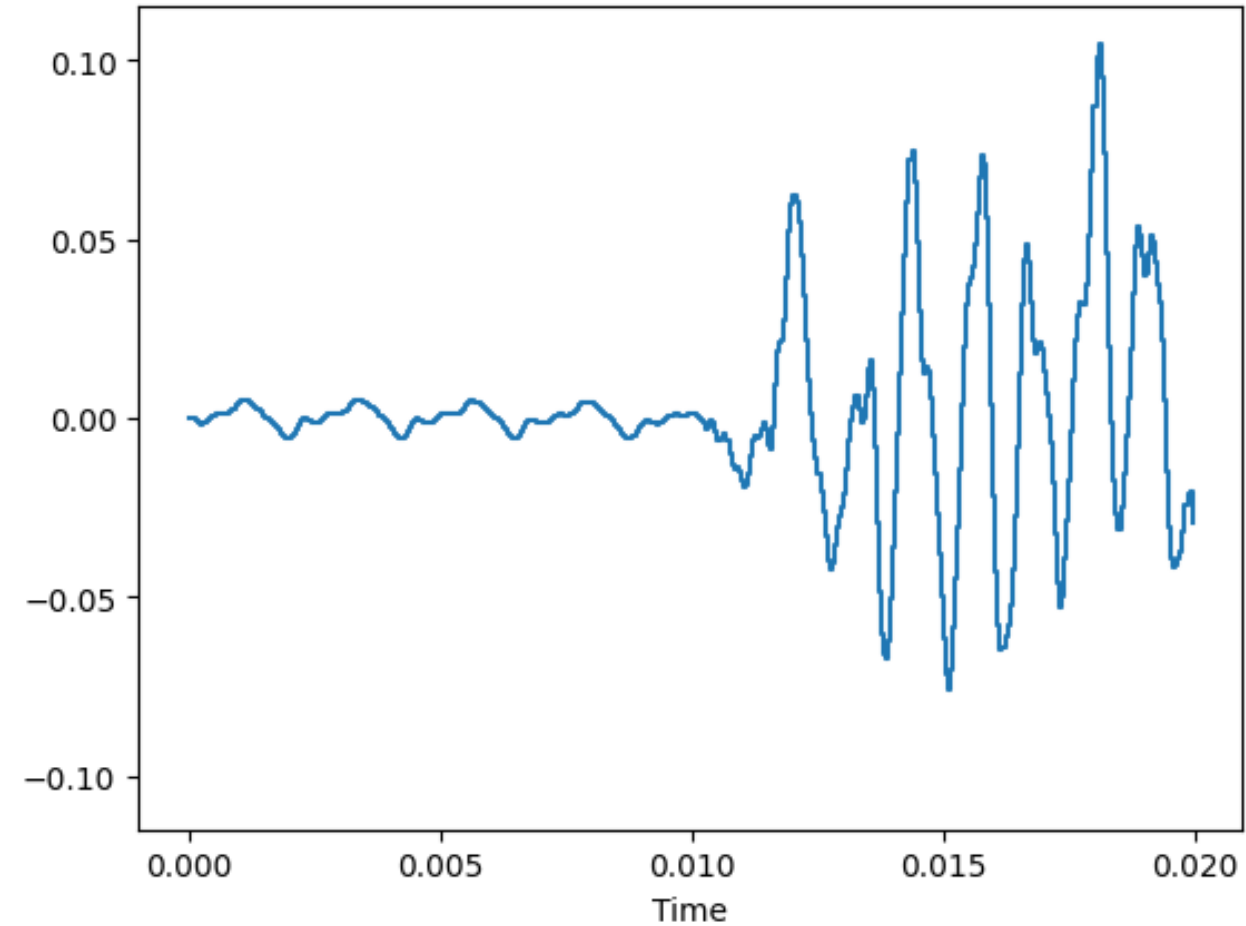
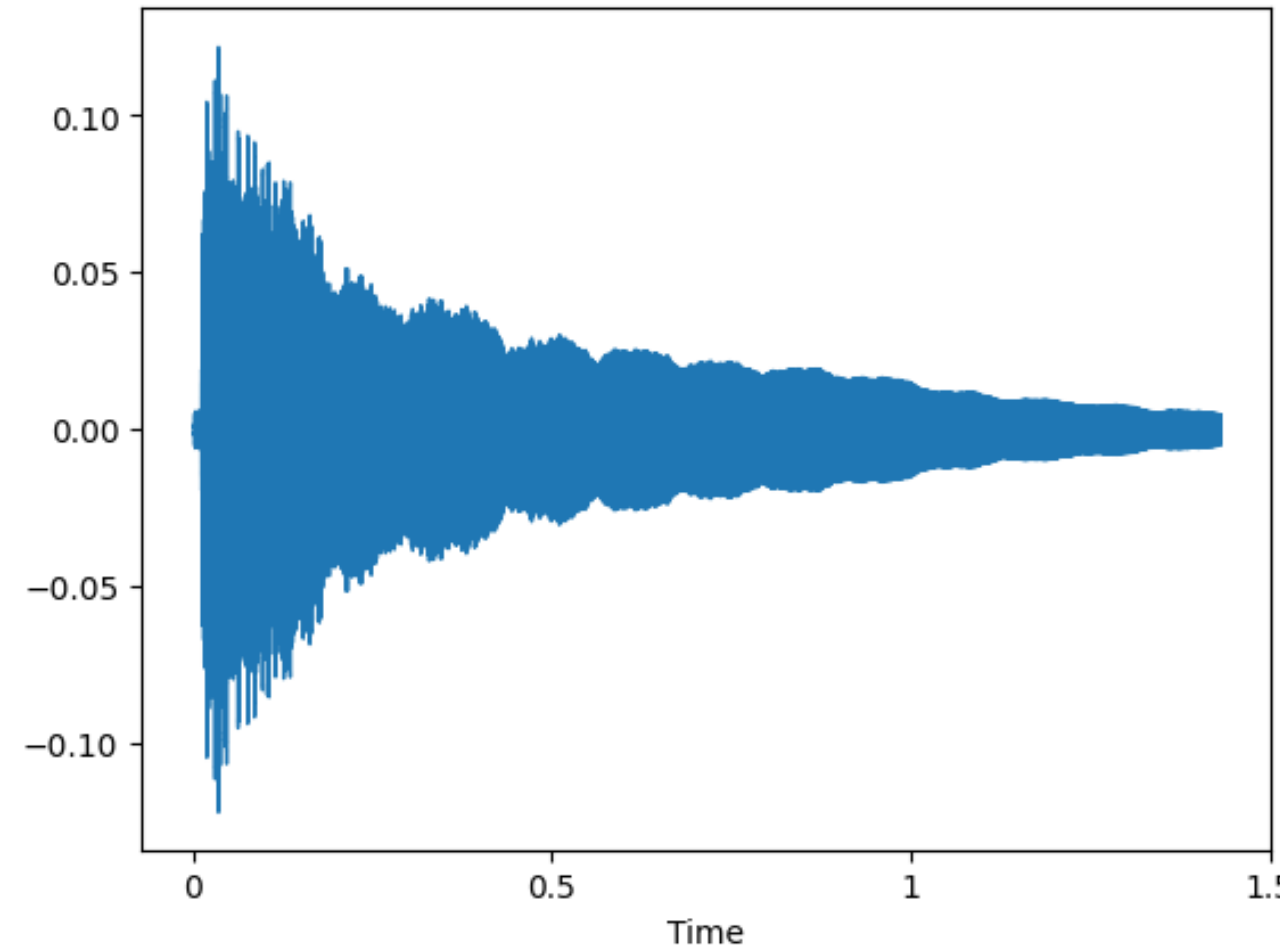
- **SAMPLING:** signal is sampled at uniform time intervals to discrete points
- **QUANTISATION:** amplitude is quantised with limited number of bits

CD

- **SAMPLE RATE:** 44.100 Hz
- **BIT DEPTH:** 16 bits



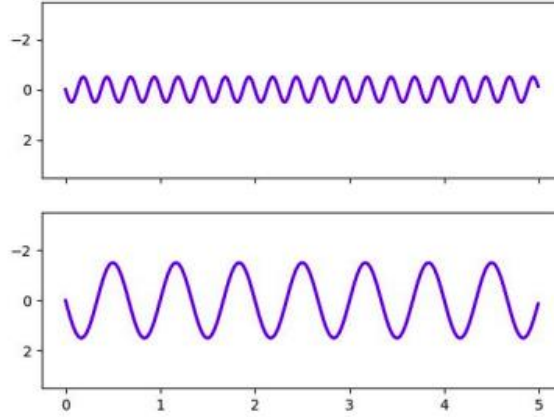
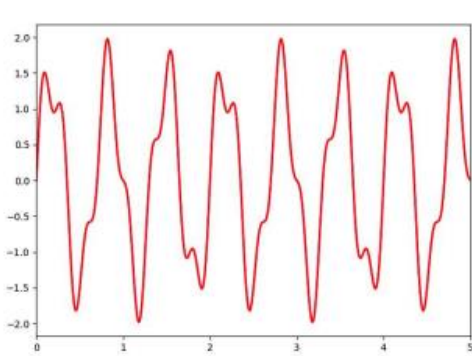
Soundwaves in the real world



Real world sound are much more complicated

Fourier Transform

Decomposing Signals into Sine Waves

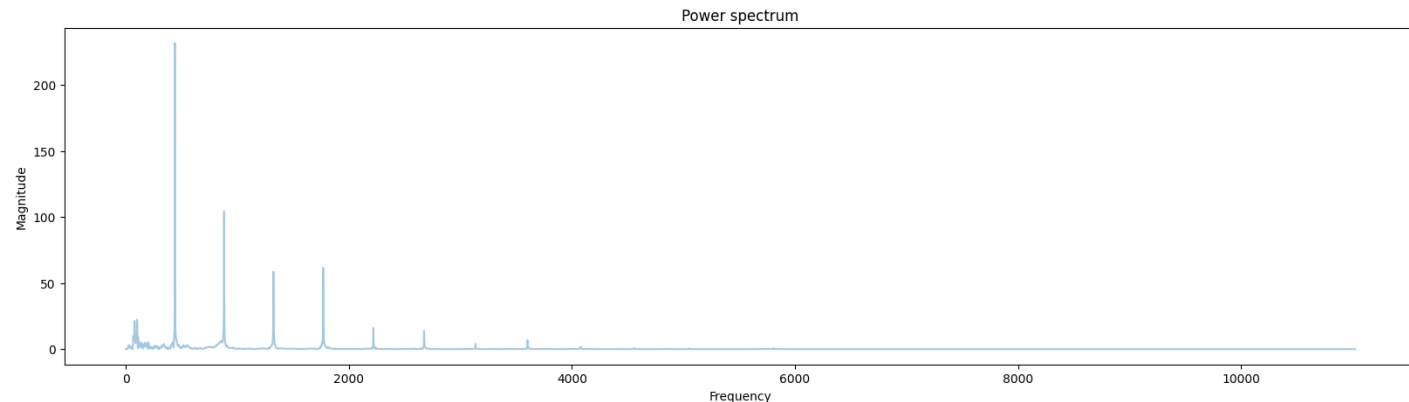


$$A_1 = 0.5, f_1 = 4, \varphi_1 = 0$$

$$A_2 = 1.5, f_2 = 1.5, \varphi_2 = 0$$

$$s = A_1 \sin(2\pi f_1 t + \varphi_1) + A_2 \sin(2\pi f_2 t + \varphi_2)$$

- Sound wave is a sum of sine waves oscillating at different frequencies
- We need to decompose complex periodic signal into its constituent sin waves (frequencies)



Fourier Transform

Decomposing Signals into Sine Waves (2)

The Fourier Transform of a signal $x(t)$ is:

$$X(f) = \int_{-\infty}^{\infty} x(t)e^{-j2\pi ft} dt$$

Where $X(f)$ is the **frequency domain representation**, showing how much of each frequency exists in the original signal.

Digital Fourier Transform (DFT)

FT for Digital Signals

The Discrete Fourier Transform (DFT) is the version of Fourier Transform used for digital (discrete) signals.

The DFT of a discrete signal $x[n]$ is:

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j2\pi kn/N}$$

Where k corresponds to discrete frequency components and N is the number of samples.

Why It's Useful

Real-world signals are discrete (e.g., digital audio). DFT helps convert a time-domain signal (samples) into its frequency-domain components.

Example Application

MP3 compression: DFT helps identify and remove inaudible frequencies to save space without losing sound quality.



Fast Fourier Transform (FFT)

FT for Digital Signals

A commonly used algorithm for computing the DFT is the fast Fourier Transform or FFT.

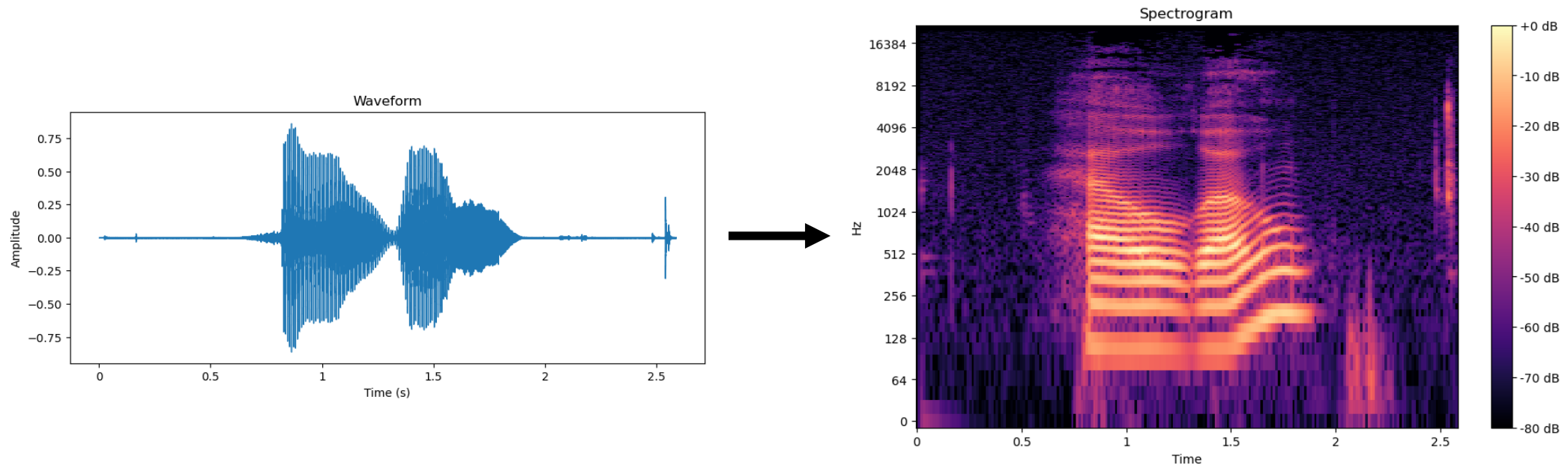
This implementation of the DFT is very efficient but only works for values of N that are powers of 2.

The **results** of the FFT tell us **the energy at each frequency band**.

N.B. Human hearing, is not equally sensitive at all frequency bands; it is less sensitive at higher frequencies.

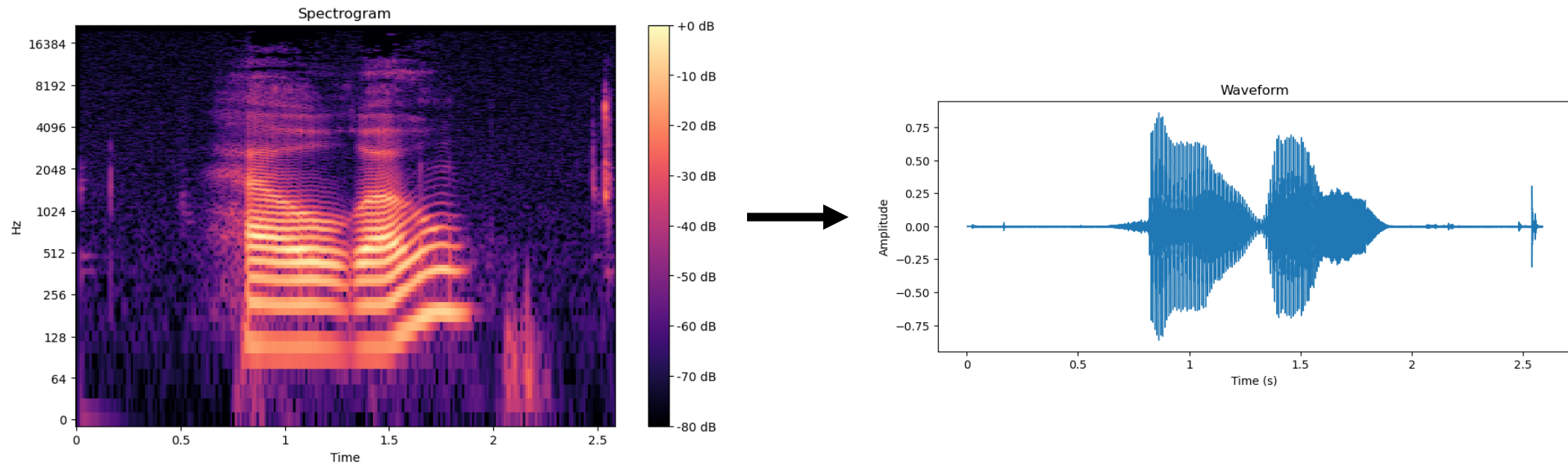


Short Time Fourier Transform (STFT)



- Computes several FFT at different intervals (or, over windows)
- Preserves time information using a fixed frame size (e.g. 2048 frames)
- Produces a **spectrogram**, a visual representation that shows how the frequencies of a signal change over time.
- Spectrogram: time x frequency x energy

Inverse Short-Time Fourier Transform (ISTFT)



- The ISTFT reverses the process of STFT, **converting the time-frequency representation back** into the original **time-domain signal**.
- This allows us to manipulate signals in the frequency domain (e.g., filtering, noise reduction) and then **recover the original signal**.

Overview of Speech Representations

We can adopt

- **Spectrogram-like representations**
 - Spectrogram
 - Mel-Spectrogram
- **Handcrafted features representations** that can be extracted from
 - Raw waveform (e.g. zero-crossing rate, energy)
 - Spectrogram (e.g. MFCCs)

Audio Features

Some examples

Feature	Waveform	Spectrogram
RMS Energy	✓	
Zero-Crossing Rate	✓	
Spectral Centroid		✓
Mel-Frequency Cepstrum Coefficients (MFCCs)		✓
Chroma Features		✓

Audio Features (2)

There is a **vast number of features** we can extract, depending on the task and domain.

For example:

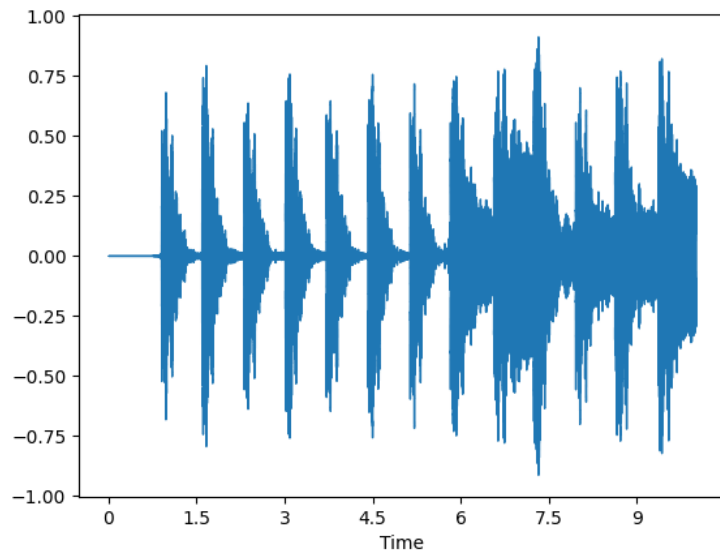
- **Prosodic Features:** Pitch, intonation, rhythm, and stress patterns (important for speech and emotion analysis).
- **Formants:** Resonant frequencies of the vocal tract (key to speech production analysis).
- **Energy and Duration Measures:** Length and loudness of sounds.
- **MFCCs and Frequency Features:** Capture rich spectral properties for tasks like speech and music classification.

Choosing the right set of features is crucial for designing effective audio processing and machine learning pipelines.

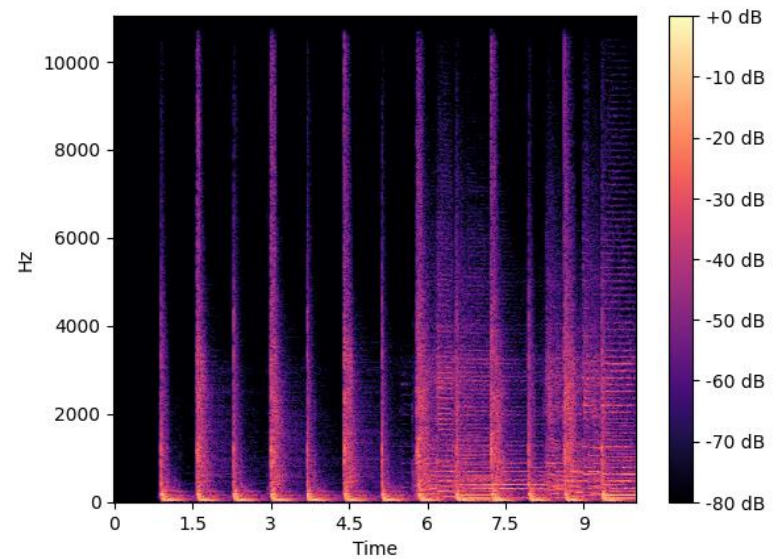


Feature Extraction

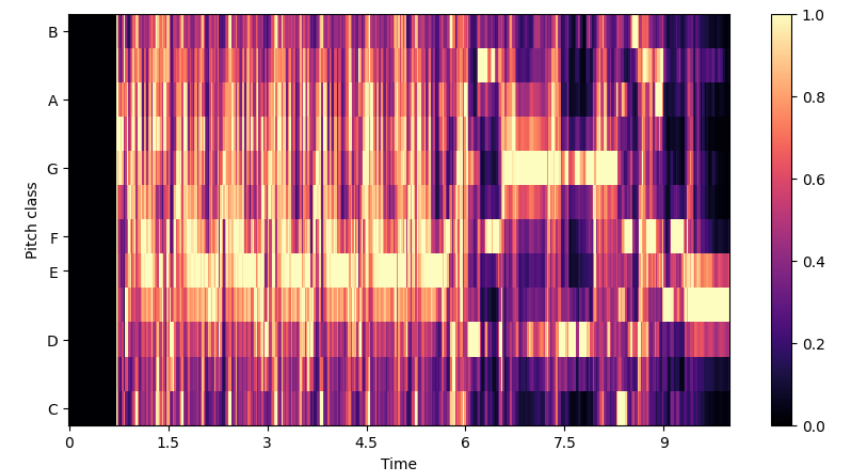
Example on “Do I Wanna Know?” by Arctic Monkeys



Waveform



Spectrogram



Chroma Features



Notebook – Part 1

Generating Spectrogram from Waveform



1. Scan QR code
2. File → Save a copy in Drive
3. Run and modify!



02

SPEECH

TASKS

What are Speech Tasks?

Automating the analysis, generation, and acquisition of **human voice**

- **Analysis:** input is speech signal, output is some **representation** that supports useful action
- **Generation:** input is that **representation**, output is speech signal
- **Acquisition:** obtaining the **representation** and necessary algorithms, from knowledge and data

Welcome

What's Natural Language Processing?



What is NLP? (more detail)

Automating the analysis, generation, and acquisition of human ("natural") language

- Analysis (or "understanding" or "processing" ...): input is language, output is some **representation** that supports useful action
- Generation: input is that **representation**, output is language
- Acquisition: obtaining the **representation** and necessary algorithms, from knowledge and data

Representation?

Paolo Torroni

Natural Language Processing

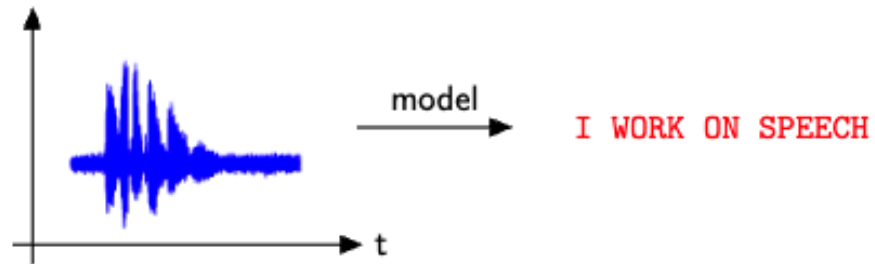
Fall 2024

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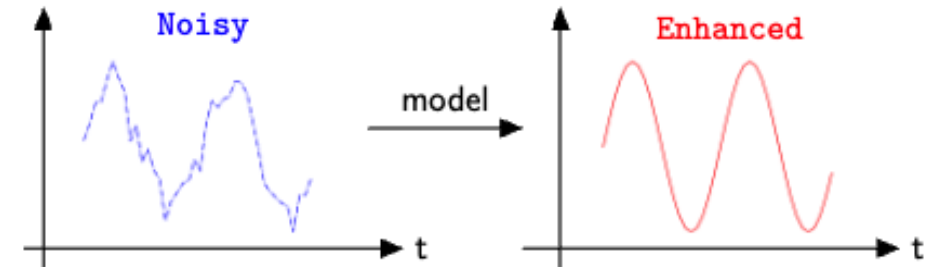


Typical Applications in Speech and Audio Modeling

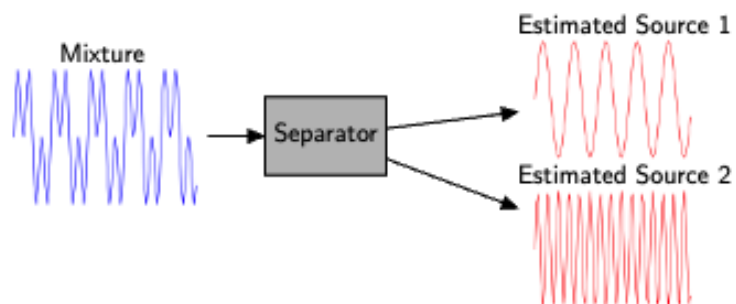
Speech Recognition (ASR)



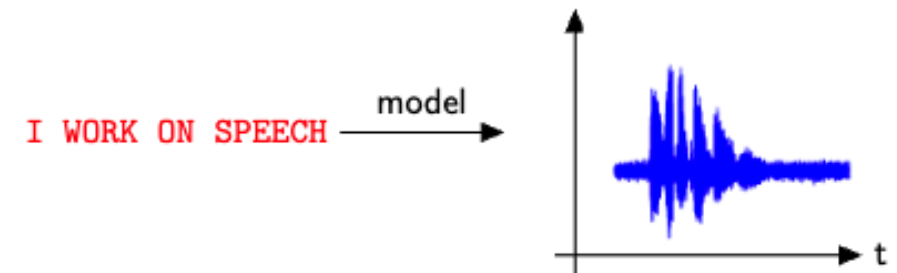
Speech Enhancement



Speech Separation

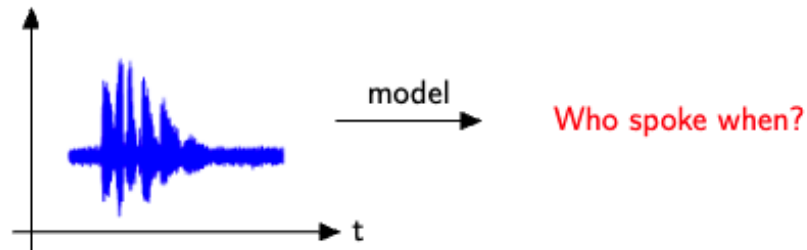


Text-to-Speech (TTS)

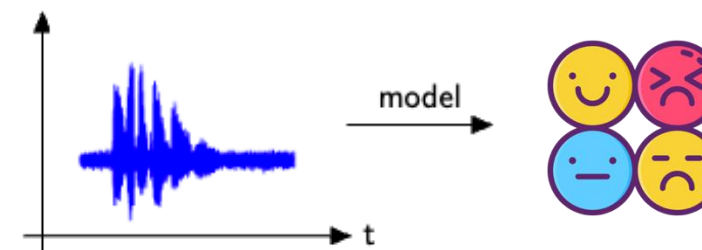


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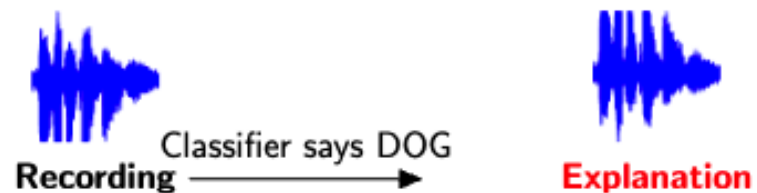
Speaker Diarization



Speech Emotion Recognition



Neural Network Explanation



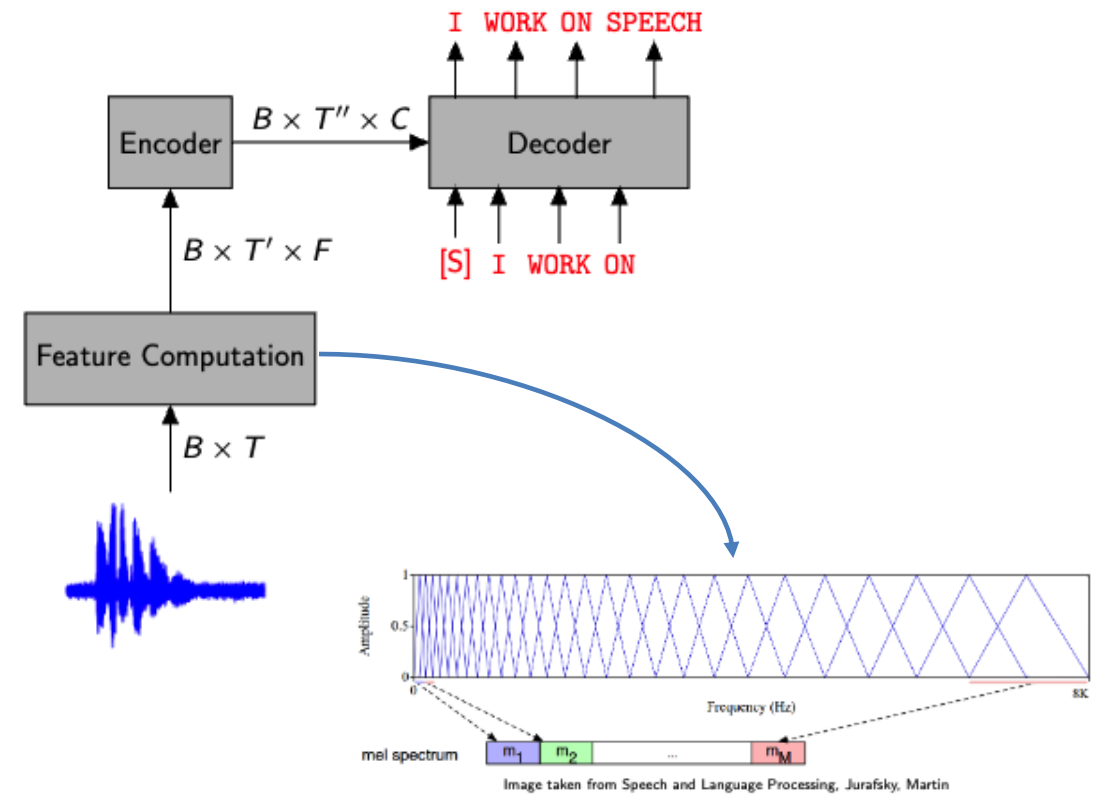
Other problems: Generating deep fakes, Detecting deep fakes, Music Source Separation, Music Transcription, Sound Event Detection/Classification...

Examples of Architectures

ASR

Encoder-Decoder Sequence-to-Sequence Learning

- The basic architecture for ASR is the **encoder-decoder** (implemented with either RNNs or Transformers).
- The encoder-decoder architecture is particularly appropriate when **input and output sequences have stark length differences**, as they do for speech, with very long acoustic feature sequences mapping to much shorter sequences of letters



Examples of Architectures

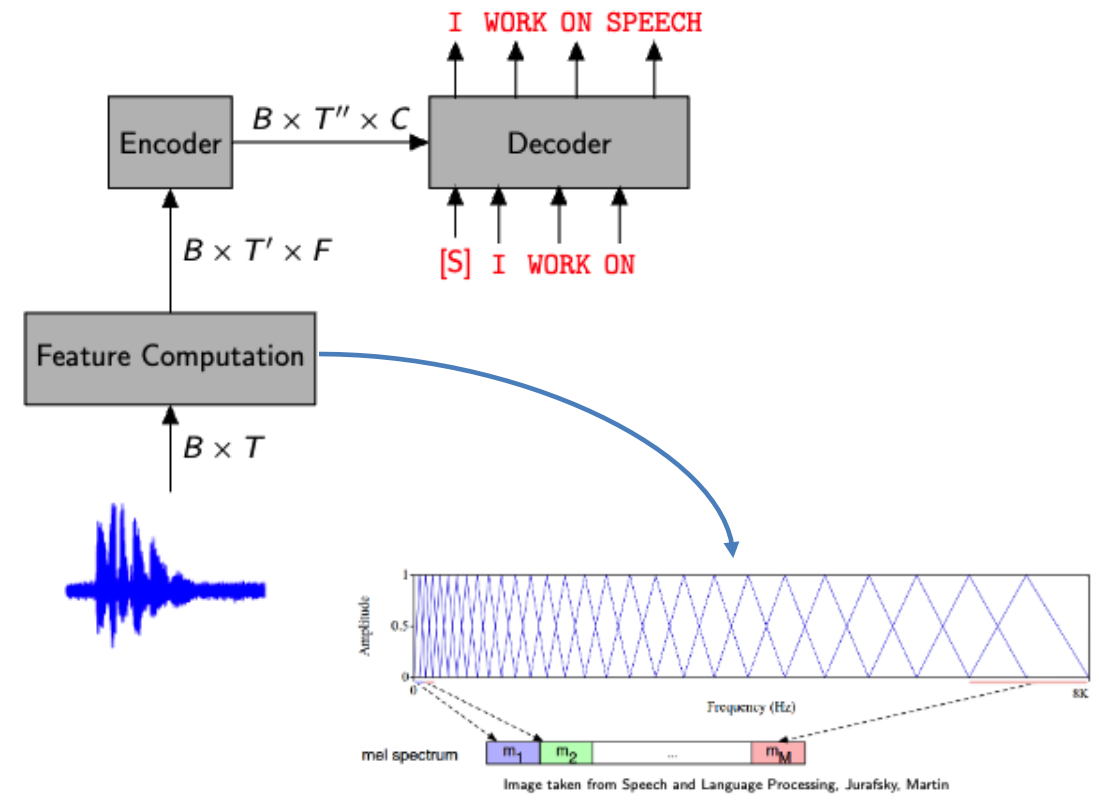
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Feature Computation

- We turn an input waveform $x \in \mathbb{R}^T$ into a time-frequency representation $x' \in \mathbb{R}^{T \times F}$
- ASR systems typically use mel-transformed spectra



Examples of Architectures

ASR

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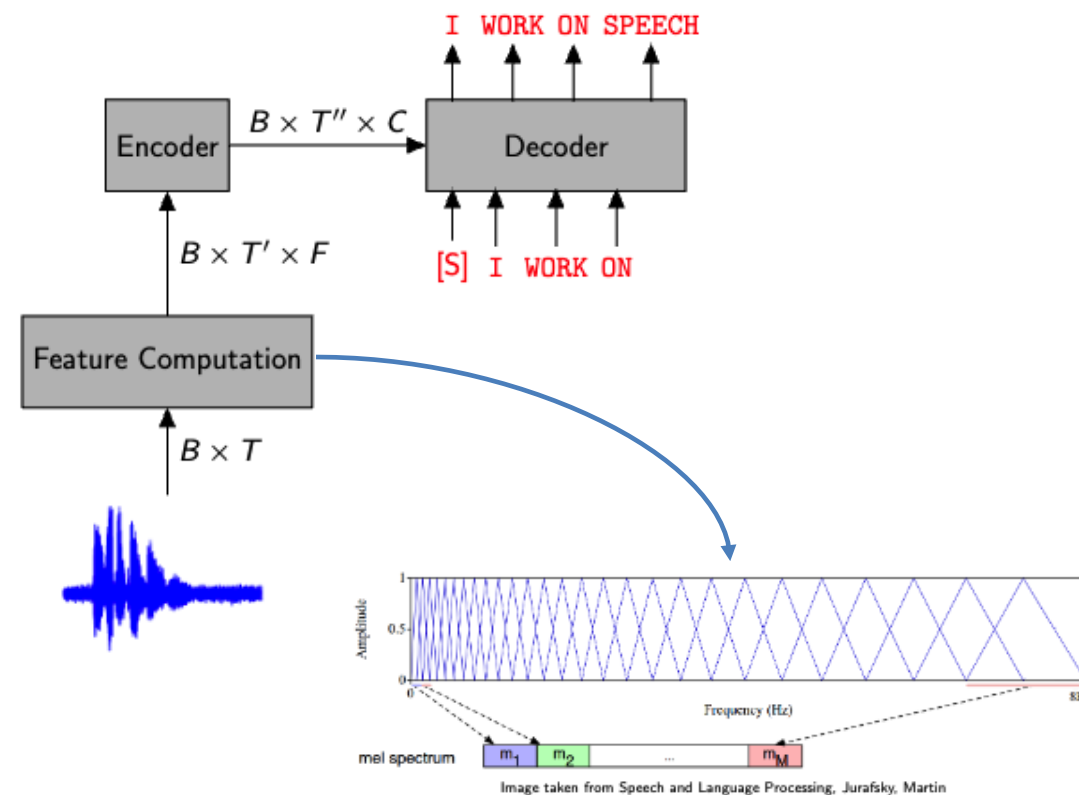
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Encoder-Decoder Model

- **Encoder:** processes speech features to extract meaningful patterns.
- **Decoder:** uses these patterns to generate predictions step by step. It takes the output vector of the encoder and creates an output sequence. The decoder will predict the next tokens of the output sequence based on the received context and its own previous predictions.



Examples of Architectures

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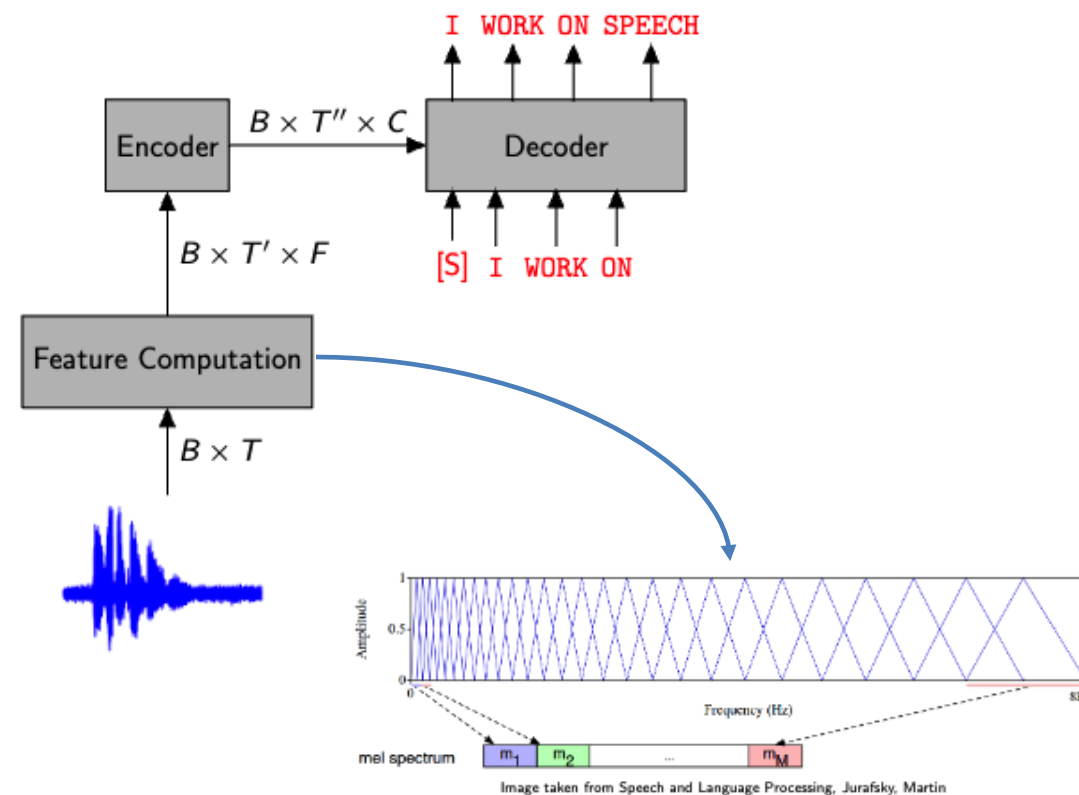
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Training

- Input: audio only
- Target transcriptions are used to compute the **loss** to guide the decoder (e.g. CTC loss, cross-entropy)



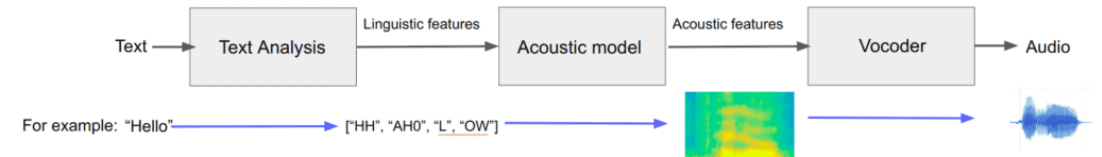
Examples of Architectures

A typical modern TTS pipeline

Again, an encoder-decoder architecture! 😊

Encoder-Decoder Model

- **Encoder:** encodes the text into a **latent representation**, capturing linguistic features like pronunciation, prosody and rhythm.
- **Decoder:** takes the encoded text representation and predicts a **mel-spectrogram**, a time-frequency representation of speech. Uses **attention mechanisms** to align text and speech for proper timing and flow.
- **Neural Vocoder:** converts the predicted mel-spectrogram into a realistic audio waveform. Common vocoders are WaveNet or HiFi-GAN.



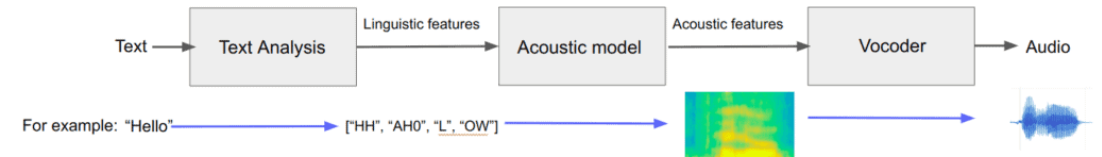
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Training

- **Input:** only text is used as input to the encoder.
- **Target Output:** speech (e.g. audio waveforms or mel-spectrograms) is used as the **ground truth** for the decoder to predict.
- **Loss Functions:** measure the difference between predicted and true mel-spectrograms (e.g., Mean Squared Error) and waveform quality.
- **Alignment Learning:** models with attention mechanisms (e.g., Tacotron) or explicit aligners (e.g., FastSpeech) learn how text corresponds to speech timing.

03

“LLMs FOR SPEECH”: SPEECH FOUNDATION MODELS



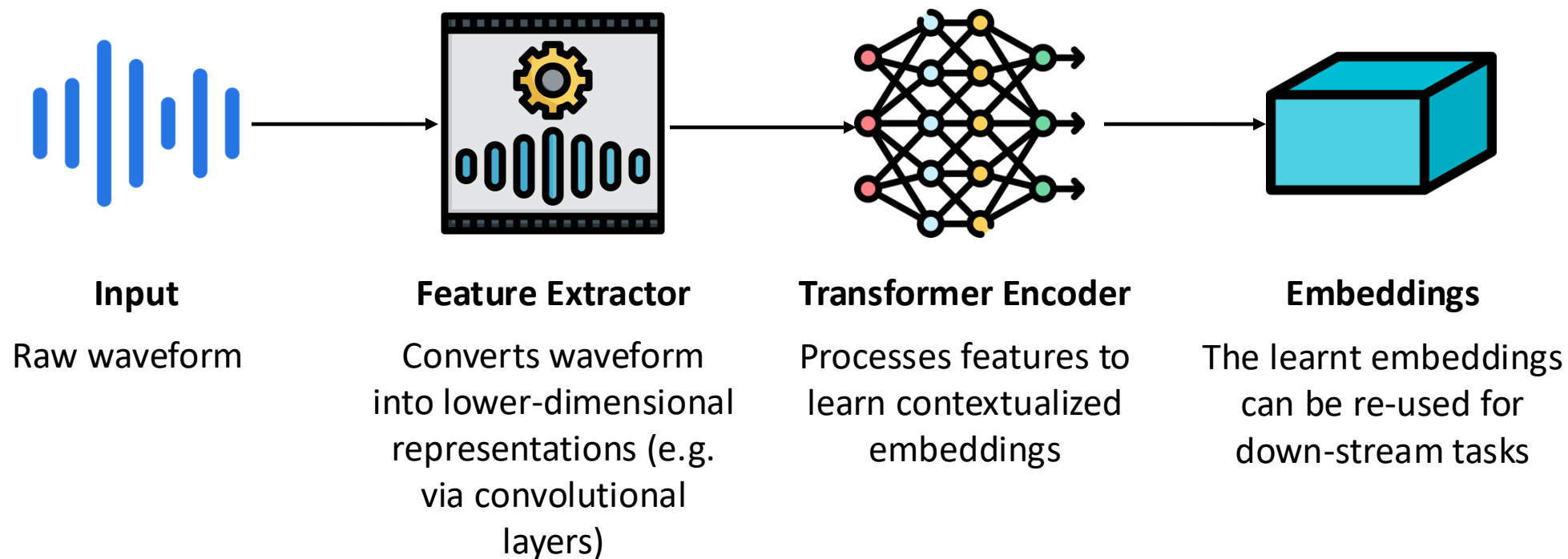
Speech Foundation Models



- **Speech Foundation Models** are **pre-trained neural networks** that learn rich audio representations from a huge amount of speech data.
- Typically, SFM are first trained in an **unsupervised** or **self-supervised manner** on large amounts of unlabeled speech data.
- As for LLMs, these models allow developing applications with minimal data specific of the downstream task, optimizing in this way the training time and the work for data collection.
- These models capture **general-purpose speech representations** that can be fine-tuned for various downstream speech-related tasks.
- They are mainly built upon **transformer-based** architectures.



Common Architecture of SF Transformer-Based Models



Why SFM are particularly useful in Speech Processing?

- Handle variability: accents, noise, speaking style
- Robust representations transferable across tasks
- Enable low-resource fine-tuning

Using SFM Effectively

- Fine-tuning: add head + train end-to-end
- Feature extraction: freeze SFM & train lightweight model
- Trade-off: compute cost vs. flexibility



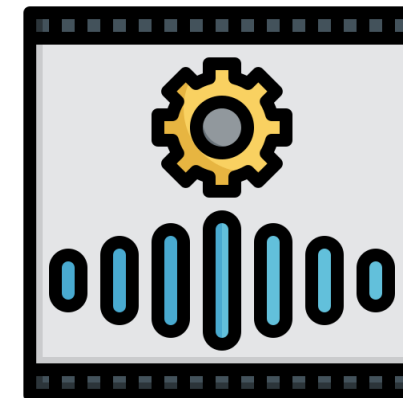
Speech and Audio downstream tasks



Fig. 5 The diagram shows how SFMs process the speech waveform into high-level representations for different speech tasks

Output Head with SFM fine tuning

a classification or regression head (one or few linear layers) is added on top of the SFM to map the speech representations into the task specific classes or regression values. This approach requires training the **output head** and **fine-tuning part** of the SFM with task-specific data



SFM as feature extractor

the representations (embeddings) produced by the SFM can be used directly as features that feed models to be trained on task specific data, or other processing algorithms.

Speech and Audio downstream tasks

Output Head with SFM Fine-Tuning

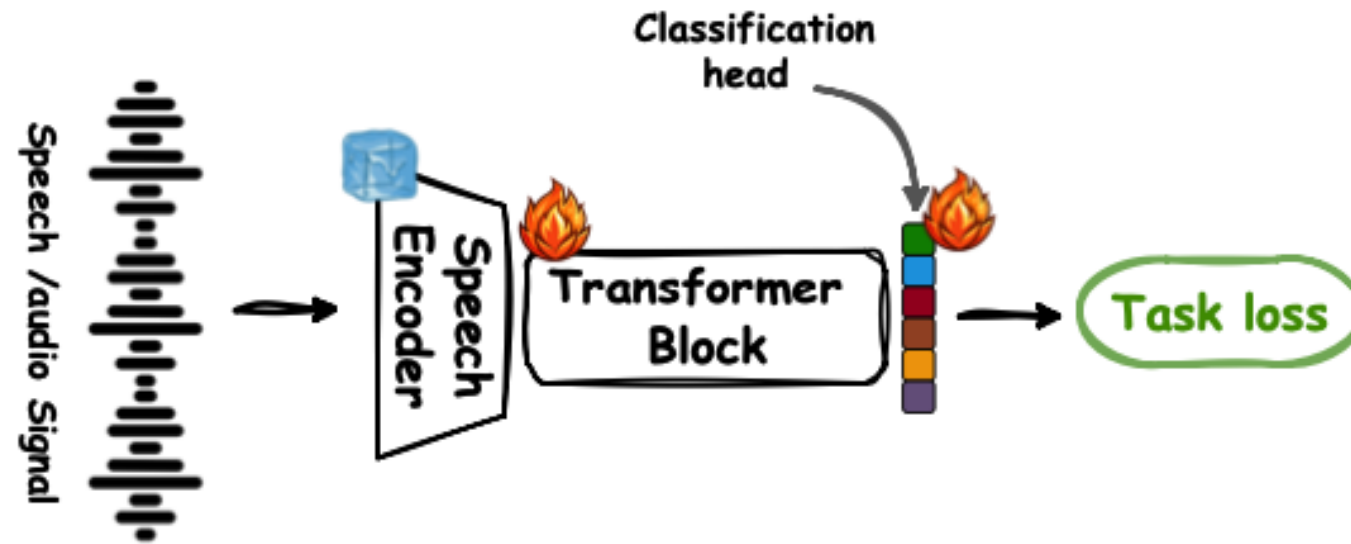


Fig. 6 Fine tuning/adaptation of an SFM with classification head using task-specific data and loss.

Speech and Audio downstream tasks

SFM as Feature Extractor (1)



Why

- Traditional speech tasks relied on handcrafted features or statistical models
- Transformers leverage massive datasets and self-supervised learning to **capture deep, contextual representations**



Key Benefits

- Extract **general-purpose embeddings** for downstream tasks (e.g., ASR, emotion recognition, speaker verification)
- Handle **raw waveforms** directly without the need for handcrafted features
- Learn from **large-scale unlabelled data**, reducing dependence on labeled datasets.

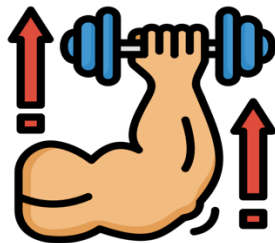


Popular Models

- **Wav2vec 2.0**: Pioneered self-supervised learning for speech
- **HuBERT**: Introduced clustering-based pretraining
- **WavLM**: Advanced speech representation learning with speech masking techniques



Speech Technology: Mission Accomplished or Still a Work in Progress?



Strengths

Robust Representations

Generalize well across accents, noise, and environments

Self-Supervised Learning

Reduce reliance on labeled data by extracting meaningful features from raw audio

Versatility

Adaptable to many tasks like ASR, speaker ID, and emotion recognition.



Open Challenges

Multimodality

Limited integration with non-audio data (e.g., text, video)

Fine-Tuning

Task-specific adaptation still requires significant resources

Extreme Variability

Challenges with rare accents, languages, and noisy audio

Handling Long Speech Sequences

Memory constraints limit processing long audio, preventing models from fully capturing context

Interpretability and Bias

Lack of transparency and risks of demographic biases



Notebook – Part 2

Accessing Speech Foundation Models with Hugging Face



1. Scan QR code
2. File → Save a copy in Drive
3. Run and modify!

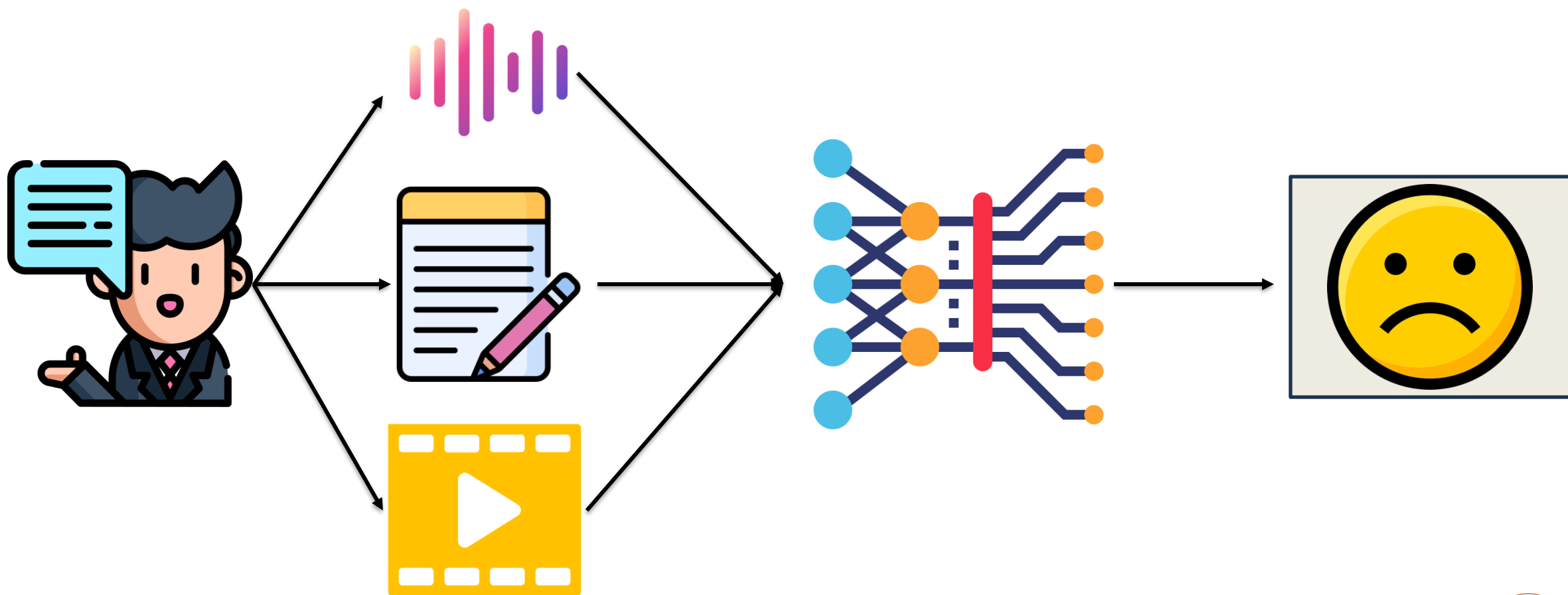


04

SPEECH AND MULTIMODALITY

Multimodal NLP

What we say and how we say it



Multimodal NLP

Addressing the Big 5 of Multimodal Learning (1)

Representation

- How to encode diverse modalities (e.g., speech, text) into a shared or complementary space.
- **Example:** Capturing both acoustic features (intonation, rhythm) and textual semantics for emotion detection embeddings.

Fusion

- How to combine information from modalities to maximize synergy
- **Example:** Integrating speech prosody and textual content for sarcasm detection
- **Note:** Representation and fusion often go hand in hand, as effective combination relies on high-quality, shared representations.



Multimodal NLP

Addressing the Big 5 of Multimodal Learning (2)

Alignment

- Linking corresponding elements across modalities in time or meaning.
- **Example:** Synchronizing speech segments with their transcriptions for automatic speech recognition (ASR).

Translation

- Mapping information between modalities to enable understanding.
- **Example:** Generating text captions from spoken language in audio description systems.

Co-Learning

- Leveraging shared information between modalities to improve learning.
- **Example:** Using text data to improve speech emotion recognition through shared embeddings.

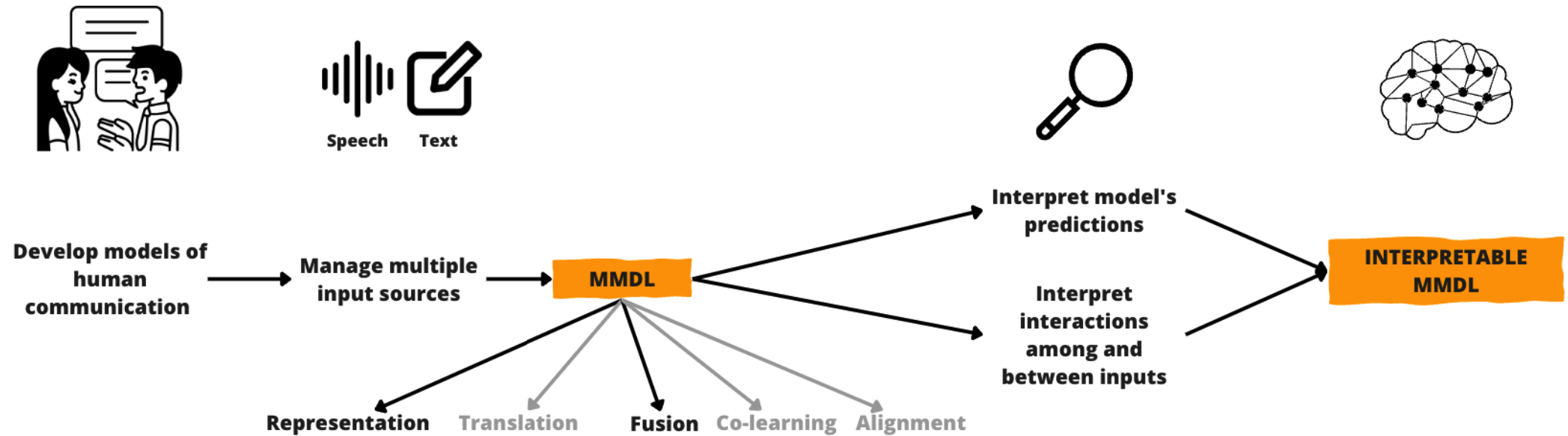


05

ABOUT MY RESEARCH IN MULTIMODAL NLP

Objective

Developing Interpretable Multimodal Neural Architectures



Method

Two Case Studies

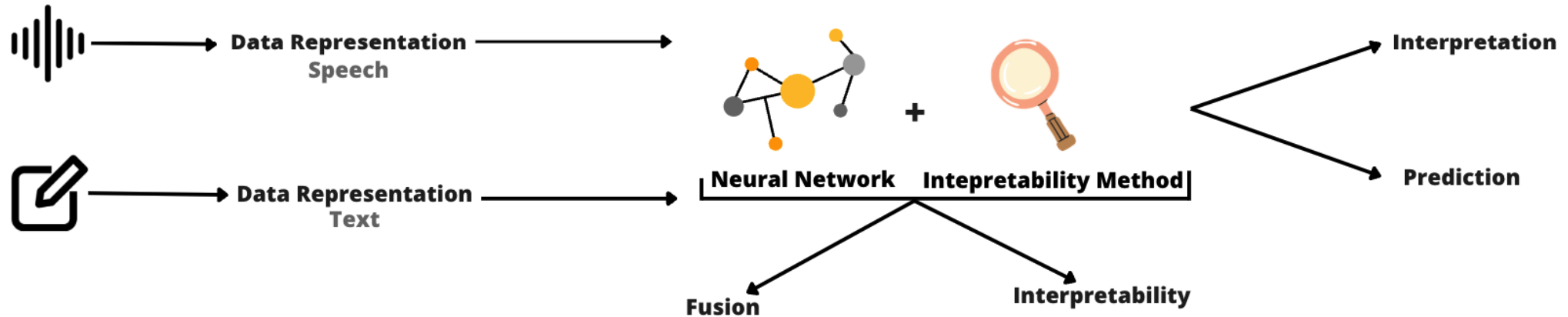


Automatic Depression
Detection from
Clinical Interviews



Multimodal Argument
Mining in Political
Debates

Approach



Rationale for Case Studies



Multimodal Argument Mining in Political Debates (MAM)

- **Current Focus:** Mostly text-based
- **Potential of Audio:** Paralinguistic features (e.g. tone, prosody) underexplored
- **Challenge:** Few multimodal datasets
- **Interpretability:** Needed to reveal how audio influences argumentative structure



Automatic Depression Detection from Clinical Interviews

- **Current Focus:** In real scenarios, clinicians assess both what is said and how it's said
- **Proven Role of Speech:** Outperforms text in automatic diagnosing of depression
- **Challenge:** Limited clinical data due to privacy and ethical concerns
- **Interpretability:** Key to understanding which speech features indicate depression, together with the speech content

Shared Themes

- **Core Modality:** Speech + Transcription
- **Low-Resource:** Scarce multimodal data in both fields
- **Generalizability:** Improving interpretability techniques adaptability across different datasets and contexts.
- **Interpretability:** Crucial for both scenarios to improve trustworthiness, predictions and annotations



Possible Impact of Multimodal Interpretable Systems



Multimodal Argument Mining in Political Debates (MAM)

- **Democratic Support:** Interpretable systems can **enhance transparency** in political discourse by analyzing both what is said and how it is said
- **Decision Support Systems:** Assists in identifying misleading arguments (e.g., fallacies) and promotes **informed decision-making** in democratic processes
- **Impact:** Helps citizens and policymakers understand argument structures, fostering **more informed** public debates



Automatic Depression Detection from Clinical Interviews

- **Decision Support Systems:** In both cases, interpretable multimodal systems enhance **transparency** and support informed decision-making
- **Real-World Integration:** Such systems can provide actionable insights for both **democratic processes** and **clinical care**
- **Clinical Process Support:** Interpretable models can integrate into **clinical workflows**, providing **decision support** for clinicians assessing depression
- **Biomarker Discovery:** Allows clinicians to identify **new speech-based biomarkers**, enhancing diagnosis and patient monitoring
- **Impact:** Improves diagnostic accuracy and fosters trust in digital tools, aiding in **personalized treatment pathways**



06

SPEECH AND AUDIO PROCESSING AROUND THE WORLD





Imperial College London



(me 😊)



ALMA MATER STUDIORUM
UNIVERSITÀ DI BOLOGNA

AND MANY MORE... 😊

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COLLABORATIONS

Research Topics

- Multimodal AI for Human Expression (speech × text × music)
- Speech & paralinguistic signal modeling
- Multimodal Argument Mining & political discourse analysis
- AI for clinical assessment & decision support
- Music Information Retrieval & creative AI
- Explainable & perceptually-grounded model interpretation



Project Proposal

- **Topic:** Producing Time-Domain Listenable Explanations for Speech Classifiers
- **People Involved:** max 2 students (a group of two)
- **Timeline:** The projects should ideally be conducted between December/January and April/May.
- If you are interested, please send me a **manifestation of interest** via email [e.mancini@unibo.it] with the following subject line: **[NLP PW] - XAI Speech - [Your Name] [Your Surname]**
- If the results are interesting, the objective is to pursue a scientific publication and/or extend this work into a thesis project.

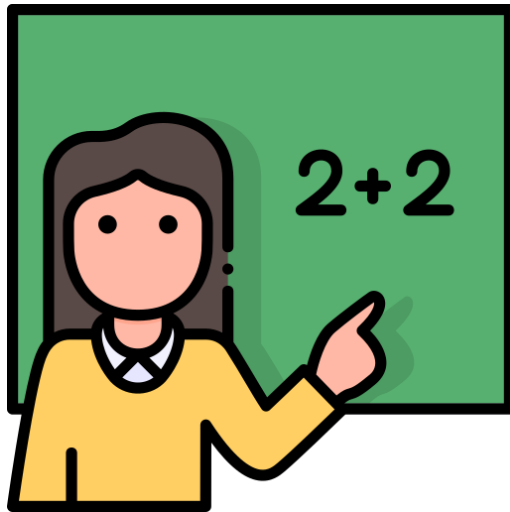


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USEFUL CONTENTS

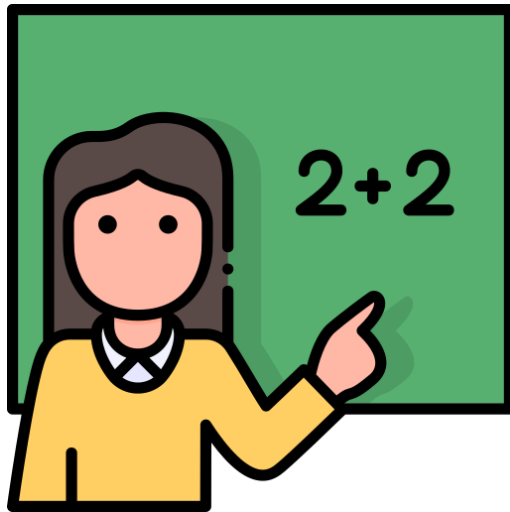
Audio Signal Processing for Machine Learning

- Coursera free course of Audio Signal Processing for Music Applications by prof. Xavier Serra (MTG-UPF, Barcelona)
- Series of lectures about Audio Signal Processing for Machine Learning by Valerio Velardo (The Sound of AI)
- Slides of the course Machine Learning for Signal Processing by prof. Cem Subakan (MILA & University of Laval, Canada – SpeechBrain Group)



Speech Processing (1)

- Slides about Speech/Audio in Machine Learning for Signal Processing Group, useful to better understand Speech Architectures (MILA & University of Laval, Canada – SpeechBrain Group)
- THE book in NLP and Speech Processing «Speech and Language Processing» (Jurafsky and Martin)



Speech Processing (2)

ASR

- Listen-Attend-Spell: <https://arxiv.org/pdf/1508.01211.pdf>
- On CTC: https://www.cs.toronto.edu/~graves/icml_2006.pdf, <https://distill.pub/2017/ctc/>

TTS

- Tacotron: <https://arxiv.org/pdf/1703.10135.pdf>
- Transformer TTS: <https://arxiv.org/pdf/1809.08895.pdf>
- Tacotron2: <https://arxiv.org/pdf/1712.05884.pdf>
- ECAPA-TDNN: <https://arxiv.org/pdf/2005.07143.pdf>

Speech Separation

- Sepformer: <https://arxiv.org/abs/2010.13154>, <https://arxiv.org/abs/2202.02884>
- REAL-M: <https://arxiv.org/pdf/2110.10812.pdf>

Speech Processing

Cross Model Representations

- CLAP: <https://arxiv.org/pdf/2206.04769.pdf>
- UI-CLAP: <https://arxiv.org/abs/2305.01864>

Interpretability for Audio

- L2I: <https://arxiv.org/pdf/2202.11479v2.pdf>
- L-MAC: <https://arxiv.org/abs/2403.13086> 127 / 129
- L-MAC TD: <https://arxiv.org/abs/2409.08655>

Some Toolkits and Libraries

- **NVIDIA NeMo** is a highly modular and scalable toolkit for building state-of-the-art speech models, with a focus on ASR, text-to-speech (TTS), NLP, and neural machine translation (MT). It is part of the NVIDIA AI ecosystem and integrates well with NVIDIA hardware for accelerated training and inference.
- **SpeechBrain** is a comprehensive, open-source toolkit developed to simplify the development of deep learning models for speech processing tasks. It aims to offer an all-in-one solution that covers various speech-related tasks such as ASR, speaker recognition, speech enhancement, and emotion recognition.
- **Fairseq**, developed by Facebook, is a general-purpose sequence-to-sequence learning toolkit that supports various tasks, including speech recognition, text translation, and text generation. Fairseq is known for its flexibility and the ability to handle diverse data modalities, including text and speech.
- Finally, **Hugging Face Transformers** is one of the most popular libraries for NLP but has recently expanded its support to include speech tasks such as ASR. The library provides easy access to pre-trained models through its Model Hub, simplifying the process of training and fine-tuning models for specific tasks.





ALMA MATER STUDIORUM
UNIVERSITÀ DI BOLOGNA

Credits:

Eleonora Mancini

Language Technologies Lab, DISI, University of Bologna

e.mancini@unibo.it

www.unibo.it